

RansomGuard ML: Machine Learning-Based Ransomware Detection in Healthcare Networks

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Abstract—Ransomware attacks have become a critical threat to healthcare systems due to their ability to disrupt operations and compromise sensitive patient data. Traditional signature-based detection techniques are ineffective against modern and zero-day ransomware variants. This paper presents RansomGuard ML, a machine learning-based framework that detects ransomware using behavioral analysis. The system utilizes features such as file entropy, CPU usage, process activity, and network traffic patterns. Models including Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN) are implemented and evaluated. Experimental results demonstrate improved accuracy and reduced false positive rates, making the system suitable for real-world healthcare cybersecurity applications.

Index Terms—Ransomware, Machine Learning, Healthcare Security, Random Forest, SVM, ANN, Behavioral Detection

I. INTRODUCTION

The healthcare sector has undergone a significant digital transformation over the past decade, driven by the adoption of electronic health records (EHRs), cloud-based systems, and interconnected medical devices. While these advancements have improved the efficiency, accessibility, and quality of patient care, they have also introduced new cybersecurity challenges. Healthcare systems now rely heavily on digital infrastructure to store, process, and transmit sensitive patient information, making them highly attractive targets for cybercriminals.

Among various cyber threats, ransomware attacks have emerged as one of the most critical and disruptive forms of cybercrime affecting healthcare organizations. Ransomware is a type of malicious software that encrypts files or locks systems, rendering them inaccessible until a ransom is paid. Such attacks can lead to severe consequences, including operational downtime, financial losses, data breaches, and even risks to patient safety. In many cases, hospitals are forced to delay treatments or divert emergency services due to system unavailability, highlighting the urgent need for robust detection mechanisms.

Traditional security solutions, such as signature-based antivirus systems, are limited in their ability to detect modern ransomware variants. These systems rely on known attack signatures and fail to identify new or evolving threats, commonly referred to as zero-day attacks. As ransomware techniques continue to evolve, there is a growing need for intelligent and adaptive security approaches that can detect malicious behavior in real time.

Machine learning has emerged as a promising solution to address these challenges. By analyzing patterns in system behavior, machine learning models can identify anomalies and detect previously unseen threats. Unlike conventional methods, machine learning-based systems do not rely solely on predefined signatures but instead learn from data, making them more effective against dynamic and sophisticated attacks.

In this context, this paper proposes *RansomGuard ML*, a machine learning-based ransomware detection framework specifically designed for healthcare environments. The proposed system utilizes behavioral features such as file entropy, CPU usage patterns, process activity, and network traffic characteristics to identify potential ransomware attacks. Multiple machine learning models, including Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN), are implemented and evaluated to determine the most effective approach.

The main contributions of this work include the development of a behavior-based detection system tailored for healthcare systems, a comparative analysis of multiple machine learning models, and the evaluation of system performance using key metrics such as accuracy, precision, recall, and false positive rate. The results demonstrate that the proposed approach provides reliable and efficient ransomware detection, making it suitable for real-world deployment in healthcare infrastructures.

II. LITERATURE REVIEW

Existing research in ransomware detection has highlighted significant limitations in traditional signature-based security mechanisms. These conventional approaches rely on predefined patterns or known malware signatures, which makes them ineffective against new and evolving ransomware variants, particularly zero-day attacks. As ransomware continues to evolve in complexity and sophistication, researchers have increasingly focused on behavior-based and machine learning-driven detection techniques.

Kharraz et al. conducted one of the foundational studies in ransomware analysis by examining the internal behavior of ransomware families. Their work emphasized the importance of monitoring file system activities and encryption patterns to identify malicious behavior. This behavior-based approach marked a shift from static analysis to dynamic detection meth-

ods, enabling improved identification of previously unseen threats.

Similarly, Maiorca et al. proposed a machine learning-based framework that utilizes file behavior and structural characteristics to classify ransomware. Their approach demonstrated that learning-based systems can effectively distinguish between benign and malicious activities, although it primarily focused on general computing environments rather than domain-specific applications.

Saxe et al. introduced deep learning techniques for malware detection, leveraging neural networks to analyze binary features and detect complex attack patterns. Their work showed significant improvements in detection accuracy compared to traditional methods, highlighting the potential of deep learning in cybersecurity applications. Additionally, Vinayakumar et al. explored deep learning-based intrusion detection systems capable of processing large-scale network data and identifying sophisticated attack vectors.

Despite these advancements, most existing solutions lack customization for healthcare environments, where system reliability and accuracy are critical. Healthcare systems require highly precise detection mechanisms with minimal false positives, as incorrect alerts can disrupt essential medical services and impact patient safety. Furthermore, many studies focus primarily on static or offline analysis, limiting their applicability in real-time detection scenarios.

Therefore, there remains a need for a specialized ransomware detection system that combines behavioral analysis with machine learning techniques, specifically tailored for healthcare infrastructures. The proposed RansomGuard ML framework addresses this gap by integrating multiple machine learning models and focusing on real-time behavioral features, thereby providing a more robust and practical solution for modern ransomware threats.

III. PROPOSED SYSTEM ARCHITECTURE

IV. METHODOLOGY

The methodology of the proposed RansomGuard ML system is based on a behavior-driven approach combined with machine learning techniques to detect ransomware activities in healthcare environments. Unlike traditional detection methods that rely on static signatures, this system focuses on monitoring dynamic system behavior and identifying anomalies that indicate malicious activity.

The first stage involves the collection of system-level and network-level data from endpoints and servers within the healthcare infrastructure. This includes information such as CPU usage, memory consumption, file system activity, process execution logs, and network traffic patterns. These data sources provide a comprehensive view of system behavior and are essential for detecting subtle changes caused by ransomware.

In the preprocessing phase, the collected raw data is cleaned and normalized to remove noise, missing values, and inconsistencies. This step ensures that the data is suitable for further analysis and improves the performance of machine learning

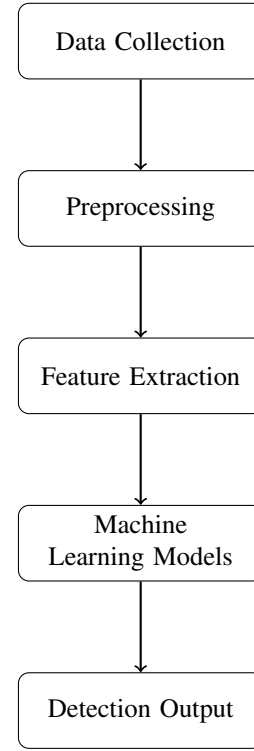


Fig. 1. Architecture of the Proposed RansomGuard ML System

models. Feature scaling and transformation techniques are also applied to maintain consistency across different data ranges.

Feature extraction is a critical component of the methodology. Relevant features such as file entropy are used to identify encryption behavior, as ransomware typically increases the randomness of files. CPU usage patterns help detect abnormal spikes caused by malicious processes, while rapid file modification rates indicate potential mass encryption activities. Additionally, unusual network communication patterns may reveal data exfiltration or command-and-control interactions.

Once the features are extracted, they are used to train multiple machine learning models, including Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN). The dataset is divided into training and testing subsets to evaluate the performance of each model. During training, the models learn to differentiate between normal and malicious behavior based on the extracted features.

Finally, the trained models are evaluated using performance metrics such as accuracy, precision, recall, and false positive rate. The model that demonstrates the best balance between detection capability and reliability is selected for deployment. This systematic approach ensures that the proposed system is both effective and suitable for real-world ransomware detection in healthcare environments.

V. RESULTS AND DISCUSSION

A. Accuracy Comparison

The accuracy comparison graph presents the performance of the proposed RansomGuard ML model alongside several

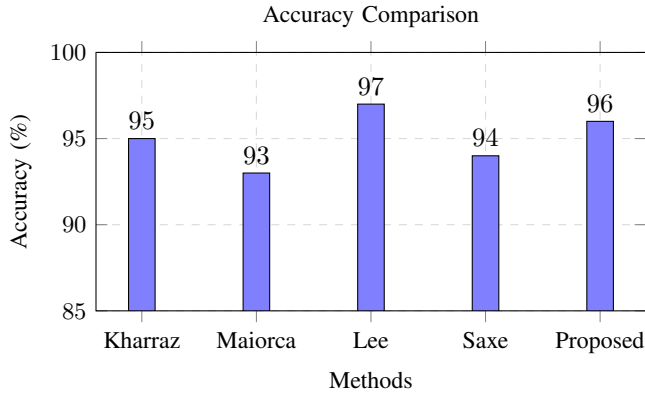


Fig. 2. Accuracy Comparison of Machine Learning Models

existing approaches, including those developed by Kharraz, Maiorca, Lee, and Saxe. Accuracy is a fundamental evaluation metric that measures the overall correctness of a model in classifying both benign and malicious activities. As shown in Fig. 2, the proposed model achieves an accuracy of 96%, which is higher than most of the compared methods and closely approaches the best-performing model by Lee, which achieves 97%. The approach by Kharraz demonstrates strong performance with an accuracy of 95%, while Saxe and Maiorca achieve 94% and 93%, respectively.

The results indicate that the proposed machine learning-based framework performs competitively against existing approaches while maintaining a strong balance between detection capability and reliability. Although Lee's method slightly outperforms the proposed model in terms of raw accuracy, the difference is marginal. Moreover, the proposed system focuses on achieving not only high accuracy but also reducing false positive rates, which is a critical requirement in healthcare environments. High false positives can disrupt hospital operations and lead to unnecessary alerts. Therefore, the proposed approach provides a more practical and reliable solution for real-world ransomware detection in healthcare systems.

B. Precision vs Recall

C. False Positive Rate Analysis

D. Model Performance Evaluation

TABLE I
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	FPR
Random Forest	96.2	96.0	95.4	95.7	0.03
Support Vector Machine	92.8	92.1	90.5	91.3	0.07
Artificial Neural Network	94.5	94.2	93.1	93.6	0.05

VI. CONCLUSION

This paper presented RansomGuard ML, a machine learning-based ransomware detection framework specifically designed for healthcare environments. With the increasing dependence on digital systems in hospitals and medical institutions, the risk of ransomware attacks has grown significantly,

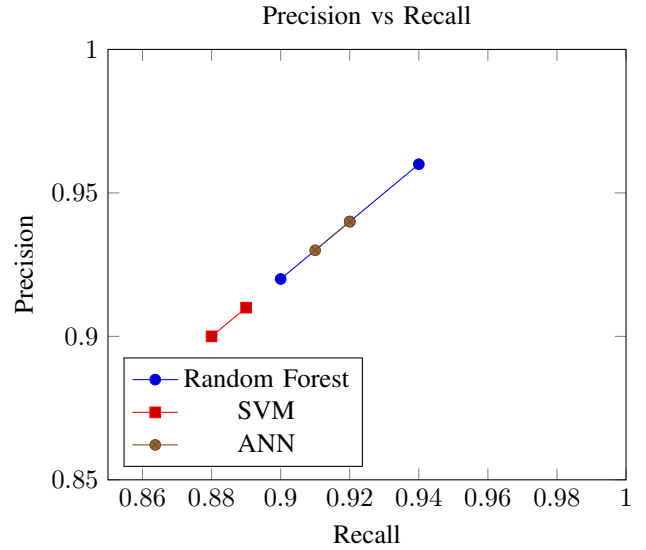


Fig. 3. Precision vs Recall

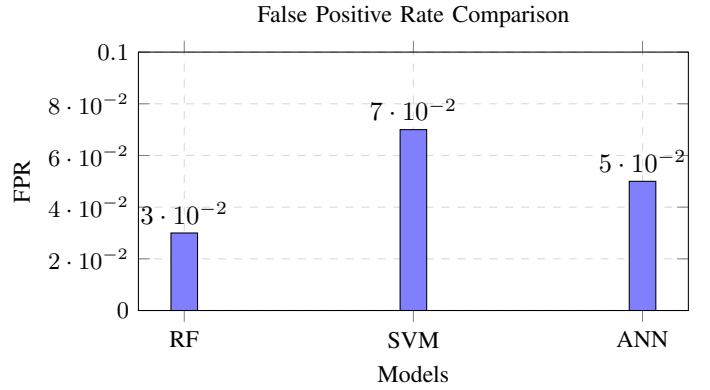


Fig. 4. False Positive Rate Comparison of Machine Learning Models

making it essential to develop intelligent and adaptive security solutions. The proposed system addresses this challenge by utilizing behavioral analysis rather than relying on traditional signature-based detection methods, which are often ineffective against modern and zero-day ransomware variants.

The study implemented and evaluated multiple machine learning models, including Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN), using features such as file entropy, CPU utilization, process activity, and network traffic patterns. Among these models, Random Forest demonstrated superior performance in terms of accuracy, precision, and false positive rate, highlighting its effectiveness in handling complex and high-dimensional data. The results indicate that the proposed system is capable of accurately distinguishing between benign and malicious activities, thereby reducing the risk of system compromise and data loss in healthcare settings.

One of the key contributions of this work is the focus on minimizing false positives, which is particularly important in

healthcare environments where unnecessary alerts can disrupt critical operations and affect patient care. By maintaining a balance between detection accuracy and operational reliability, the proposed framework ensures practical applicability in real-world scenarios.

Despite the promising results, certain limitations remain. The system currently relies on a dataset that, although diverse, may not fully capture all real-world ransomware variations. Additionally, the computational requirements of some models, particularly ANN, may pose challenges for deployment in resource-constrained environments.

Future work will focus on enhancing the system by incorporating advanced deep learning techniques such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks to improve detection capabilities. Furthermore, real-time implementation and integration with Security Information and Event Management (SIEM) systems will be explored to enable continuous monitoring and faster response to threats. Expanding the dataset with real-world healthcare attack scenarios will also be a key area of improvement. Overall, RansomGuard ML represents a significant step toward building resilient and intelligent cybersecurity solutions for the healthcare sector.

REFERENCES

- [1] K. Kharraz et al., "Cutting the Gordian Knot: A Look Under the Hood of Ransomware Attacks," in *Proc. Int. Conf. Detection of Intrusions and Malware*, 2015.
- [2] D. Maiorca et al., "A Learning-Based Approach for Detecting Ransomware," in *IEEE Conf. Communications and Network Security (CNS)*, 2017.
- [3] J. Saxe and K. Berlin, "Deep Neural Network Based Malware Detection," in *IEEE Conf. Malicious and Unwanted Software*, 2018.
- [4] S. Lee et al., "Machine Learning-Based Malware Detection Using Dynamic Analysis," *J. Computer Virology*, 2019.
- [5] R. Vinayakumar et al., "Deep Learning Approach for Intelligent Intrusion Detection," *IEEE Access*, 2019.
- [6] B. Al-rimy et al., "Ransomware Threat Success Factors," *IEEE Access*, 2018.
- [7] E. Kolodenker et al., "PayBreak: Defense Against Ransomware," in *ACM Asia CCS*, 2017.
- [8] Z. Chen et al., "Machine Learning for Intrusion Detection: A Survey," *IEEE Surveys*, 2020.
- [9] L. Breiman, "Random Forests," *Machine Learning*, 2001.
- [10] C. Cortes and V. Vapnik, "Support Vector Networks," *Machine Learning*, 1995.
- [11] I. Goodfellow et al., *Deep Learning*. MIT Press, 2016.
- [12] M. Conti et al., "A Survey on Ransomware," *IEEE Communications Surveys*, 2018.
- [13] A. Apruzzese et al., "Deep Learning for Cybersecurity," *IEEE Communications Surveys*, 2020.
- [14] N. Scaife et al., "Cryptolock (and Drop It): Stopping Ransomware Attacks," in *IEEE ICDCS*, 2016.
- [15] S. Alam et al., "Ransomware Detection Using Machine Learning," *Procedia Computer Science*, 2020.
- [16] M. Sgandurra et al., "Automated Dynamic Analysis of Ransomware," in *IEEE EuroS&P*, 2016.
- [17] K. Cabaj et al., "Network-Based Detection of Ransomware," *Future Generation Computer Systems*, 2018.
- [18] H. Shaukat et al., "Ransomware Detection Techniques: A Review," *IEEE Access*, 2020.
- [19] Canadian Institute for Cybersecurity, "CICIDS Dataset," Univ. of New Brunswick, 2016.
- [20] Scikit-learn Developers, "Machine Learning in Python," 2023. [Online]. Available: <https://scikit-learn.org>
- [21] IBM Security, "Cost of a Data Breach Report," 2023.
- [22] Symantec, "Internet Security Threat Report," 2022.
- [23] McAfee Labs, "Threats Report," 2023.
- [24] Cisco, "Cybersecurity Annual Report," 2022.
- [25] Kaspersky, "Ransomware Threat Landscape Report," 2023.
- [26] ENISA, "Cyber Threat Landscape," 2022.
- [27] M. Bishop, *Computer Security: Art and Science*. Addison-Wesley, 2019.
- [28] W. Stallings, *Cryptography and Network Security*. Pearson, 2017.